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Effect of equivalence ratios on the power, combustion stability and NO_x controlling strategy for the turbocharged hydrogen engine at low engine speeds

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ABSTRACT

Increase the equivalence ratio is a good way to improve performance of turbocharged hydrogen engines at low engine speeds. To explore the feasibility of this strategy, this paper investigated the experimental data of a 2.3 L turbocharged port fuel injection (PFI) hydrogen engine at 1500 rpm and 2000 rpm. The results showed that the power can increase from 6.8 kW to 21 kW at 2000 rpm and from 6.4 kW to 16.5 kW at 1500 rpm with increasing of the equivalence ratio. However, the equivalence ratio corroding to the biggest power is 0.8 at 1500 rpm and 0.9 at 2000 rpm because the turbocharged pressure and the volumetric efficiency at 2000 rpm are higher than the ones at 1500 rpm. The biggest BTE can reach to 30.1% at 2000 rpm and 29.3% at 1500 rpm within the range of 0.65–0.8. The covariance of indicated mean effective pressure (CoV_{imep}) of turbocharged hydrogen is lower than 1.5% at low engine speeds and the combustion stability increased with the increase of equivalence ratio. The NO_x can be reduced from 877 ppm to 0 ppm at 1500 rpm and from 1259 ppm to 17 ppm at 2000 rpm, which means the reduction efficiency of H_2+TWC can exceed 99%.

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Introduction

Global warming is the most serious environmental issue in the world and the biggest challenge for traditional energy. The carbon dioxide (CO_2) is the main part of the greenhouse and most of the CO_2 comes from the consumption of traditional energy [1]. To control the effect of greenhouse, the target of COP21 meeting is to keep global warming below 2 °C in Pairs in 2015. The internal combustion engine is an important way of energy consumption and many countries begin to control the CO_2 emission through the fuel consumption regulations. For example, the target of American is 4.32 L/km before 2025, Europe is 3.8 L/km before 2020, Japan is 4.92 L/km before 2020, China is 5 L/km before 2020 [2]. These regulations promote the development of internal combustion engines into two directions: on the one hand, the development of cleaner and more efficient conventional fuel internal combustion engines; on the other hand, the use of alternative fuels. Hydrogen has the characteristics of fast combustion speed, low ignition

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energy, wide flammable range and the combustion products without carbon, which make it become an ideal fuel for the internal combustion engines [3–5].

The mixture formation of a hydrogen fuel internal combustion engine is generally divided into two types: PFI and direct injection (DI). The advantage of DI is that the gaseous fuel is injected after the intake valve is closed, which does not occupy the cylinder volume, so that the power of the hydrogen internal combustion engine is close to that of the same displacement gasoline engine. Thomas Wallner et al. conducted tests in a 0.5 L, DI hydrogen engine and the results showed it can maintain the same power as comparable gasoline engines [6]. Alan Welch et al. compared the brake mean effective pressure (BMEP) of PFI hydrogen engine, gasoline engine and DI hydrogen engine and they found that the theoretical BMEP of DI is 1.17 times that of the gasoline engine [7]. Another advantage is that DI hydrogen engine is benefit to the combustion because it can achieve stratified combustion through the multiple injection strategy [8–10]. However, the way of DI has two problems, which limit the utilizing of DI. On the one hand, the nozzle reliability is low because the hydrogen nozzle cannot be cooled and lubricated like the oil nozzle, which will increase the possibility of the burnt out or deformation for the nozzle [11]. On the other hand, the hydrogen nozzle needs higher injection pressure than the other fuel because of minimal of hydrogen, which leads to the poor sealing. For example, Kimitaka Yamane et al. used a new material of O ring with the hardness of HS 94 with no backup ring and they found the hydrogen nozzle can operate 700 h for the durability test and 100 h for the DI hydrogen engine [12]. The nozzle of PFI can increase the reliability because of the lower injection pressure and the lower temperature than the DI nozzle. The way PFI can avoid the two problems.

However, the power of PFI hydrogen engine is less than the comparable gasoline engine because the fuel will occupy the cylinder volume and the heat value of hydrogen-air at stoichiometric ratio is 3.18 MJ/m^3, which is smaller than the 3.81 MJ/m^3 of gasoline-air. Thus, many researchers explored the effect of boosting the intake air pressure on the performance of PFI hydrogen engine. Ford Groups investigated two kinds of PFI boosting hydrogen engines and their results showed that the torque can be equal to the one of the compressed natural gas engines at the same volume [13,14]. They also introduced a 4.0 L PFI turbocharged hydrogen engine and the maximum BTE can reach to 35.5% [15]. BMW Groups reported a 0.5 L PFI hydrogen engine and the intake air pressure were boosted to 0.185 MPa [16]. They showed that the BMEP can reach to 1.8 MPa at all working conditions through adjusting the equivalence ratio. Verhelst et al. developed a 7.4 L PFI supercharged hydrogen engine [17]. They found the maximum torque can increase 60% than the one of naturally aspirated and the power can increase 30% than the one of the basic gasoline engine. Moreover, they found the way of stoichiometric ratio adding with exhaust gas recirculation (EGR) can get higher BTE than the way of lean equivalence ratio [18]. Das et al. [19] presented a 2.5 L turbocharged hydrogen engine with an intercooler, whose power can reach to 66.7 kW at 3200 rpm. Boosting pressure didn't only increase the power, the NO_x emission also increased. Mover, the torque at low engine speed and high load also decreased because the exhaust gas cannot afford enough energy to make the turbine work, which will decrease the intake air mass. To improve the performance at low engine speeds, increasing equivalence ratio is an important way because the high equivalence ratio can increase the energy into the cylinder [20].

Moreover, cycle variation is another important parameter to evaluate the performance of hydrogen engine. Kim et al. studied the effects of engine speed, hydrogen injection timing, equivalence ratio and ignition timing on the cyclic variation of DI hydrogen internal combustion engine. The results showed that COV_{imep} is suitable for the cyclic variation of characterization of combustion in DI hydrogen engine [21]. Subramanian et al. explored the cycle variation evaluation parameters of PFI hydrogen engine and found that increasing the equivalence ratio can reduce the cyclic variation [22]. Similar test conclusions were also obtained under lean conditions by Ji Changwei et al. [23]. Chen Yu studied the evaluation method of the cycle variation of hydrogen engine at the idle condition and analyzed the correlation between the maximum cylinder pressure and the corresponding three values of the crank angle with the average effective pressure. The results showed that the COV_{imep} is an ideal choose under idle condition [24]. In the past work of our research group, we have conducted the test at a 2.0 L naturally aspirated PFI hydrogen engine and found that the COV_{imep} is the best evaluation parameter to the PFI hydrogen engine at all working conditions [25]. However, the cycle variation of turbocharged PFI hydrogen at low engine speeds is different the above results because the turbine will increase the back pressure, which will deteriorate exchanging process and increase the exhaust gas residual coefficient. The result is that the cycle variation will increase at low engine speeds for turbocharged PFI hydrogen engine. Thus, it is necessary to explore the effect of equivalence ratio on the cycle variation and propose strategy to increase the combustion stability.

The NO_x emission of turbocharged hydrogen engine will increase because the high combustion temperature. The results of reference [19] showed that the NO_x emission was over 800 ppm though the equivalence ratio was controlled smaller than 0.6. Verhelst et al. proposed the solutions of EGR + TWC to control the NO_x emission and the result showed the TWC conversion efficiencies can achieve more than 99% [18]. Ford group also adopted the solution of EGR + TWC [26]. Sierens et al. compared the lean and cold EGR ways on the CFR engine [27]. They found that EGR reduced the pumping loss at low load and the heat transfer loss also decreased. These solutions can decrease the NO_x emission but there are two problems: 1) the heat capacity of water is smaller than the one of CO_2, which will drop the effect of EGR on the NOx; 2) the cooled EGR will product liquid water, which will corrode the EGR system; 3) the high temperature exhaust gas increase the possibility of the abnormal combustion-backfire. Thus, it is necessary to explore new strategy to solve the NO_x emission for turbocharged PFI hydrogen engine under the premise that other performance can meet the requirements.

This paper proposes increasing the equivalence ratio to improve the performance of turbocharged hydrogen engine at low engine speeds. To explore the effect of increasing equivalence ratio on the performance, Section Effect of equivalence ratio on the power and BTE at low engine speed shows the

power and BTE and Section Effect of equivalence ratio on the combustion stability at low engine speed focuses on the cycle variation. Hydrogen is the best reducing agent in the nature and this paper proposed using hydrogen to reduce the NO_x in the TWC. Section Effect of equivalence ratio on the NO_x and controlling strategy at low engine speed showed the NO_x emission at front and behind of TWC and the strategy to improve the performance at low engine speeds.

Material and method

A 2.3 L, 4 St, PFI hydrogen engine with a turbocharger was installed into a hydrogen engine test bench. The parameters of the turbocharged hydrogen engine are shown as Table 1 and the sketch is described as Fig. 1.

This paper mainly explored the effect of equivalence ratio (λ) on the power, BTE and NO_x controlling strategy for the turbocharged hydrogen engine at low engine speeds, so different working conditions of two engine speeds (1500 rpm and 2000 rpm) and $\lambda = 0.4–1.1$ were conducted in the test bench. To absorb and motor the turbocharged hydrogen engine, a CW250 eddy current dynamometer was installed. The oil and water cooling system, intercooler system were also installed to control the temperature of the hydrogen engine. The electric control unit (ECU) was based on the PROtronic system and the Simulink mode of the Matlab. The controlling strategies can be achieved by the ECU system. The data measure system mainly included the air flow meter, hydrogen flow meter (CMF010 and CMF025), temperature (Pt100 and K type thermocouple), pressure of inlet and outlet pipe (steady state and transient pressure sensors), environment temperature and pressure, the concentration and component of the exhaust gas (AVL DiGas 4000 Light). Moreover, a combustion analyzer of Kistler was also used in the test to measure the combustion process at different working conditions. Before conducting the test, the hydrogen engine should be warmed up to the suitable temperature, which the oil temperature was 90 °C and the temperature of the cooling water was 80 °C. Due to the limitation of the mass of hydrogen nozzle and knock [26], the throttle angle was set to 17deg at 1500 rpm and 16deg at 2000 rpm. The NO_x emission at front and behind of TWC was measured at the same working condition using the AVL DiGas 4000 Light.

Table 1 – Parameters of the 2.3 L turbocharged hydrogen engine.

Engine type	Hydrogen engine
Bore	90 mm
Stroke	90 mm
Cylinders	4
Crank radius	45 mm
Connecting rod radius	153 mm
Compression ratio	9.3
Exhaust valve open	53.6°BBDC
Exhaust valve close	44.4°ATDC
Inlet valve open	23.3°BTDC
Inlet valve close	58.7°ABDC
Maximum torque @ 3000 rpm	161 N.m
Maximum power @ 4000 rpm	59.4 kW

Results and discussions

Effect of equivalence ratio on the power and BTE at low engine speed

Equivalence ratio can increase the combustion velocity, which has effect on the power and BTE. Fig. 2 shows the power ascends with increasing of the equivalence ratio at 1500 rpm and 2000 rpm. As the equivalence ratio changes from 0.4 to 0.9, the power can increase from 6.8 kW to 21 kW at 2000 rpm while the power decreases after $\lambda > 0.9$. The reason can be explained by the descending of air mass flow when the equivalence ratio is bigger than 0.9 as Fig. 3 shown. With the increase of the equivalence ratio, hydrogen occupies more cylinder volume, which leads to the decrease of the air flow mass when the throttle opening angle keeps constant. Compared the curve of 2000 rpm with 1500 rpm in Fig. 2, the maximum of the air flow mass achieves at $\lambda \approx 0.8$ at 1500 rpm. The difference between 1500 rpm and 2000 rpm can be found in Fig. 4. The turbocharged pressure at 2000 rpm is bigger than the one at 1500 rpm, which will raise the volumetric efficiency and lead to the increase of the air flow mass.

The effect of equivalence ratio on the BTE can be found as Fig. 5 shown. BTE increases when $\lambda < 0.8$ and decreases when $\lambda > 0.8$ at 1500 rpm and 2000 rpm. The reason can be explained from two aspects: on the one hand, the hydrogen flow mass increases from 0.75 kg/h to 1.92 kg/h at 1500 rpm and from 0.85 kg/h to 2.49 kg/h at 2000 rpm as Fig. 6 shown, which means more energy into the cylinder and the higher load. The larger load can increase the mechanical efficiency as Fig. 7 shown, which the mechanical efficiency increases from 73% to 92% at 1500 rpm and from 63% to 92%. The higher mechanical efficiency means the larger BTE. On the other hand, the combustion velocity ascends rapidly with increase of the equivalence ratio, which shortens combustion duration. As Fig. 8 shown, the combustion duration (10%–90%) decreases from 25.4°CA to 5.1°CA at 1500 rpm and from 25.4°CA to 6.1°CA at 2000 rpm. These are the two factors that induce the increase of BTE when $\lambda < 0.8$. When the equivalence ratio is larger than 0.8, the effect of combustion velocity and loads decreases and more energy is taken away by the exhaust gas and cooling water [18]. The result leads to the decrease of the BTE. Moreover, the indicated thermal efficiency (ITE) decreases from 35.3% to 27.4% at 1500 rpm and from 36.7% to 27.2% at 2000 rpm as Fig. 8 shown. Thus, increasing the equivalence ratio can raise the torque and increase the BTE within a certain equivalence ratio ranges as Fig. 9 shown.

Effect of equivalence ratio on the combustion stability at low engine speed

Cycle variation is an important characteristic for the spark ignition (SI) engines and it can reflect the combustion stability. There are four methods to analyze the cycle variation of SI engines: exhaust gas composition correlation analysis method, burning flame correlation analysis method, in-cylinder pressure correlation analysis method and combustion rate correlation analysis method. Limited by the test and measurement conditions, the in-cylinder pressure correlation

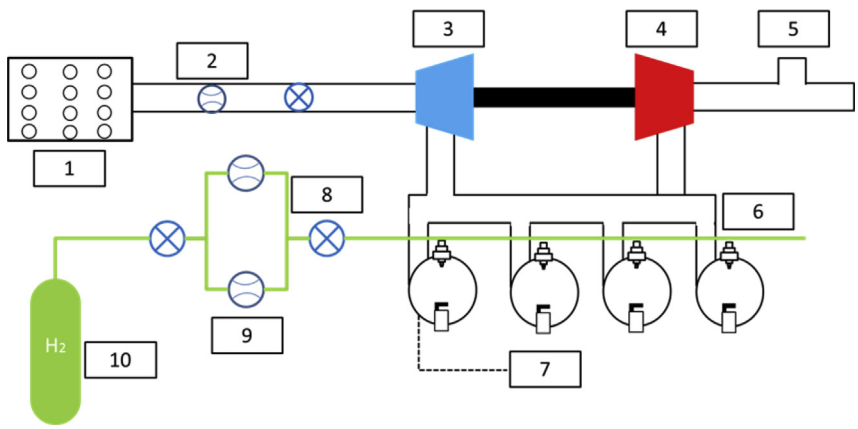


Fig. 1 – Sketch of the turbocharged hydrogen internal combustion engine (1-Air filter, 2- Air mass flow meter, 3-Compressor, 4-Turbine, 5-Exhaust analyzer, 6-Hydrogen pipeline, 7-Combustion analyzer, 8-Vavle, 9- Hydrogen mass flow meter, 10-Hydrogen bottles).

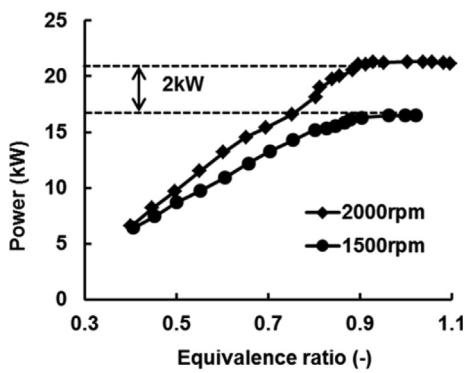


Fig. 2 – Power changes with equivalence ratio at 1500 rpm and 2000 rpm.

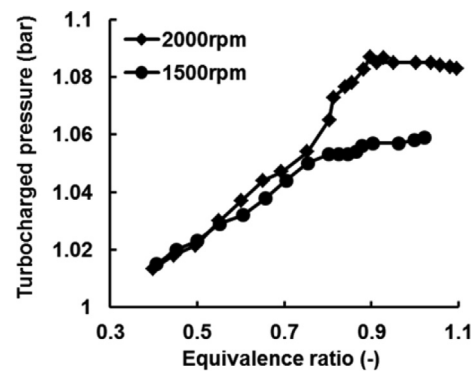


Fig. 4 – Turbocharged pressure changes with equivalence ratio at 1500 rpm and 2000 rpm.

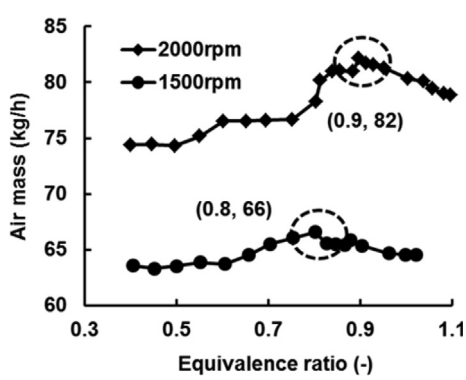


Fig. 3 – Air mass changes with equivalence ratio at 1500 rpm and 2000 rpm.

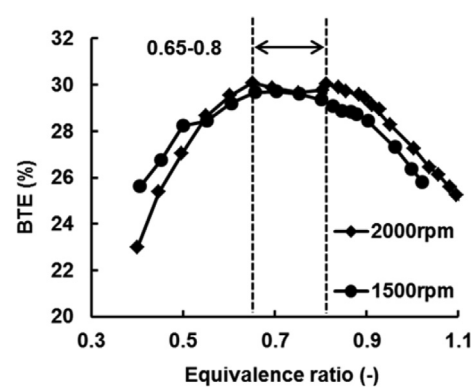


Fig. 5 – BTE changes with equivalence ratio at 1500 rpm and 2000 rpm.

analysis method has been more widely used in the study of cycle variation for SI engines. This paper also adopts this method to study the cycle variation of the turbocharged hydrogen engine. Choosing the right number of data points to study the cycle variation can reduce experimental error and test workloads. Based on the review of reference [25], this paper chooses the working condition at 2000 rpm, $\lambda = 0.4$, 0°CA, throttle opening angle (Th) = 16deg to study the right number of data points as Fig. 10 shown. The IMEP of this working condition ranges from 2.6bar to 2.8bar and the average is around 2.69bar. Fig. 11 shows the cumulative average of IMEP changes with cycle number, which the maximum fluctuation amplitude can be reach to 0.83%. When the cycle number is bigger than 120, the cumulative average of IMEP gradually approaches to the average. The fluctuation amplitude is less than 0.3% when the cycle number is larger

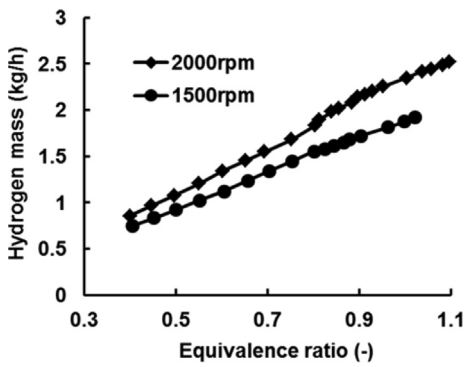


Fig. 6 – Hydrogen mass changes with equivalence ratio at 1500 rpm and 2000 rpm.

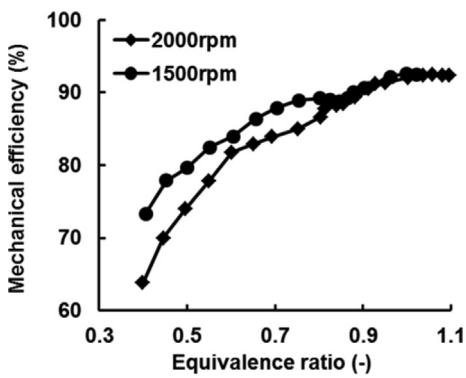


Fig. 7 – Mechanical efficiency changes with equivalence ratio at 1500 rpm and 2000 rpm.

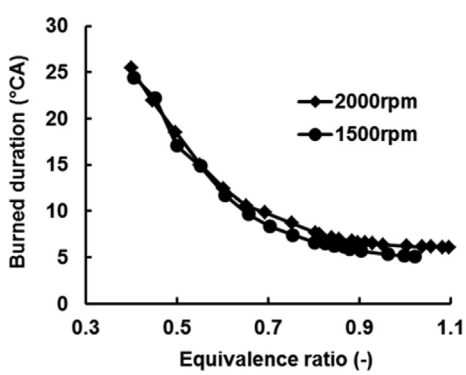


Fig. 8 – Burned duration changes with equivalence ratio at 1500 rpm and 2000 rpm.

than 150. Thus, choosing 150 continuous cycles is enough to study the cycle variation.

With the equivalence ratio changing from 0.4 to 1.0, the COV_{imep} decreases from 0.85% to 0.35% at 1500 rpm and from 1.3% to 0.4% at 2000 rpm as Fig. 12 shown. The reason can be described as following: First, the combustion velocity increases rapidly with increasing of equivalence ratio. The faster combustion velocity means the shorter combustion duration as Fig. 8 shown, which will decrease the COV_{imep}. Second, increasing the equivalence ratio can rise the intake air pressure as Fig. 4 shown, which will improve the gas exchange

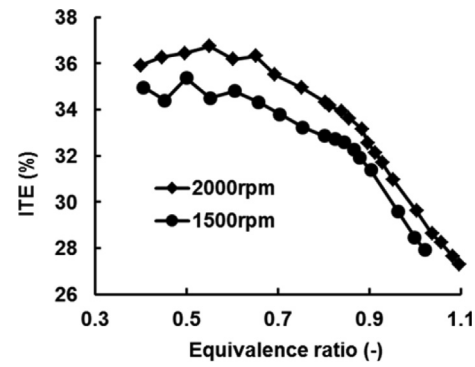


Fig. 9 – ITE changes with equivalence ratio at 1500 rpm and 2000 rpm.

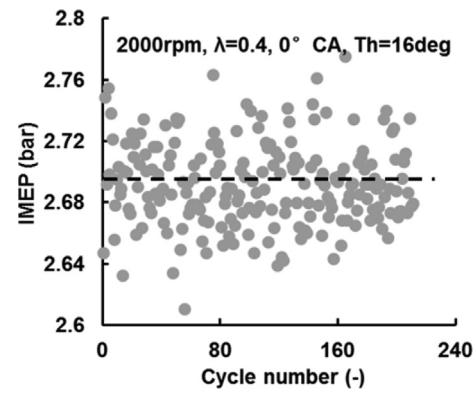


Fig. 10 – IMEP changes with cycle number at 2000 rpm, λ = 0.4, 0°CA, Th = 16deg.

process and increase the volumetric efficiency. The better gas exchange process and the higher volumetric efficiency will decrease the cycle variation of the gas composition in the cylinder. Finally, the pumping mean effective pressure (PMEP) decreases from −0.38bar to −0.29 bar at 1500 rpm and from −0.46bar to −0.34 bar at 2000 rpm with increase of the equivalence ratio as Fig. 13 shown. The decrease of PMEP will descend the exhaust gas residual coefficient, which will increase the combustion stability of the mixture in the cylinder. The effect of above three factors leads to the results of the cycle variation. The COV_{imep} of turbocharged hydrogen is

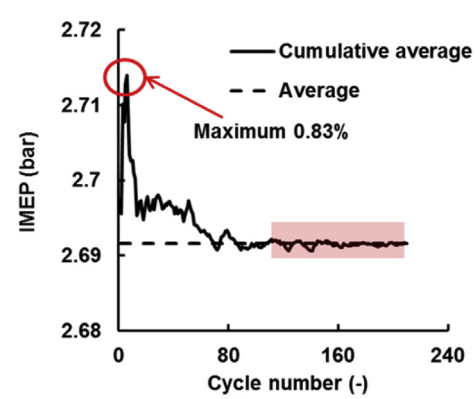


Fig. 11 – Cumulative average and average change with cycle number at 2000 rpm.

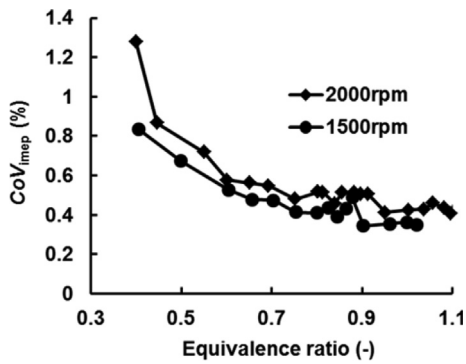


Fig. 12 – COV$_{imep}$ changes with equivalence ratio at 1500 rpm and 2000 rpm.

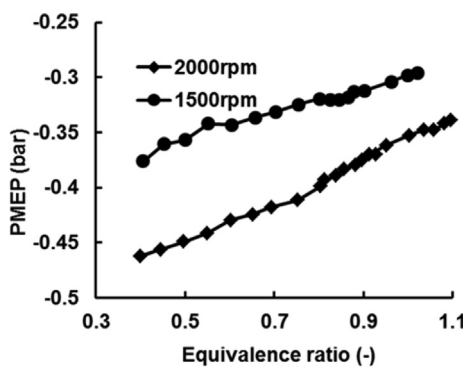


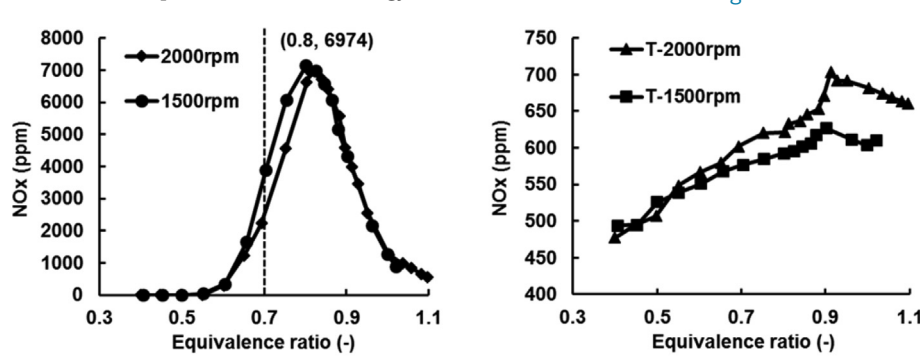
Fig. 13 – PMEP changes with equivalence ratio at 1500 rpm and 2000 rpm.

lower than 1.5%, which is a very small value to the SI engines. Thus, increasing the equivalence ratio can descend the cycle variation and ascend the combustion stability for the turbocharged hydrogen engine at low engine speeds.

Effect of equivalence ratio on the NO_x and controlling strategy at low engine speed

Increasing power and reducing NO_x emission are the mainly contradictions for the hydrogen engines [28,29]. Increasing equivalence ratio can raise the power and BTE at low engine speeds, so this section will explore the effect of equivalence ratio on the NO_x emission and put forward a strategy to improve the comprehensive performance of turbocharged hydrogen engine at low engine speeds. In the pure hydrogen internal combustion engines, the main component of NO_x is NO and its formation satisfies the Zeldovich mechanism: temperature, oxygen concentration and action time. The tendency of NO_x emission changing with equivalence ratio has been described in the reference [30], so this paper introduce the difference at low engine speeds. As Fig. 14 shown, when the equivalence ratio increases from 0.6 to 0.8, the NO_x emission at 1500 rpm is higher than the one at 2000 rpm. The reason is that the oxygen concentration and combustion temperature are roughly equal but the action time at 1500 rpm is longer than the one at 2000 rpm. The temperature in the TWC also is shown as Fig. 14(b).

Hydrogen is the best reductant and many researchers have focused on the utilizing it to reduce the NO_x emission such as H_2-SCR. Hydrogen as the fuel for the hydrogen engines can be gained more easily than the other fuel engines, which can remove the equipment of producing or carrying hydrogen. Moreover, TWC is the cheapest and most conventional catalytic device on modern vehicles. Thus, this paper puts forward the strategy of reducing NO_x emission using H_2-TWC and the equivalence ratio of slightly higher than stoichiometric ratio. In order to explore the feasibility of the strategy, the NO_x emission at front and behind of TWC changing with equivalence ratios has been showed as Fig. 15 and Fig. 16 at 1500 rpm, Th = 17deg, MBT and 2000 rpm, Th = 16deg, MBT. The NO_x emission at behind of TWC is less than the one at front of TWC when the equivalence ratio is smaller than 0.95 at 1500 rpm and 2000 rpm. There are two factors to explain this phenomenon: on the one hand, few lubricating oil will be burned at operating the hydrogen engines, which produce the HC and CO. These HC and CO can reduce the NO_x emission in the TWC. On the other hand, there is still a small amount of hydrogen in the exhaust gas when the equivalence ratio is smaller than 1, which has been proved by the Ford groups [17]. Thus, the HC, CO and H_2 can reduce the NO_x emission in the TWC even the exhaust gas has high oxygen concentration but the amount of reduced NO_x emission is a bit little due to few reductant and low conversion efficiency. The NO_x emission behind TWC rapidly decrease to zero when $\lambda > 0.95$. The reason is that the oxygen concentration declines to zero and more H_2 can be used to reduce the NO_x emission in the TWC. The comparison of NO_x emission between front and behind of TWC is showed as Fig. 17 when the equivalence ratio is



(a) (b)

Fig. 14 – NO_x and temperatures in the TWC change with equivalence ratio at 1500 rpm and 2000 rpm.

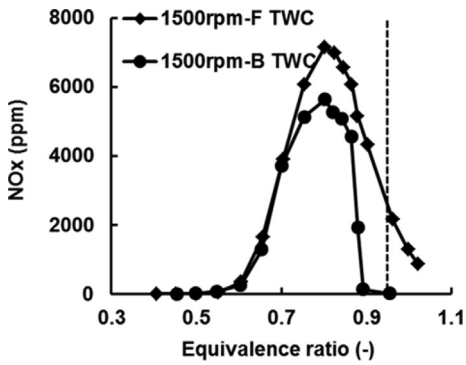


Fig. 15 – NO_x at front and behind of TWC changes with equivalence ratio at 1500 rpm.

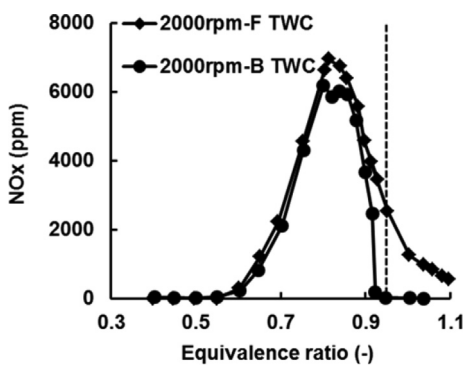


Fig. 16 – NO_x at front and behind of TWC changes with equivalence ratio at 2000 rpm.

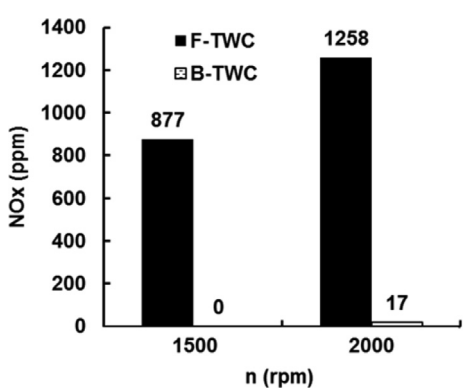


Fig. 17 – Compared the NO_x at front TWC with behind of TWC at 1500 rpm and 2000 rpm.

stoichiometric ratio. The conversion efficiency of NO_x emission in the TWC is more than 99%, which means the NO_x emission is close to zero. Thus, using H_2 as reductant to reduce the NO_x emission in the TWC is a feasible strategy.

Conclusions

Improving the performance at low engine speeds for the PFI turbocharged hydrogen engine means increasing the power, enhancing BTE and controlling the NO_x emission. This paper puts forward the strategy of H_2+TWC and the equivalence ratio slightly higher than stoichiometric ratio to improve the comprehensive performance. Based on the above results, some conclusions can be got as following.

(1) The hydrogen flow mass can block the increase of air flow mass at high equivalence ratio when the throttle opening angle keeps constant for PFI hydrogen engines because the total flow mass into the cylinder is constant. The equivalence ratio corroding to the maximum power is 0.8 at 1500 rpm and 0.9 at 2000 rpm, which is affected by the volumetric efficiency and turbocharged pressure. Thus, choosing the suit equivalence ratio can increase both the power and BTE when the volumetric efficiency is constant.
(2) Increasing equivalence ratio cannot only ascend the power, but the BTE also increases within a certain equivalence ratio ranges (such as 0.6–0.8) because the equivalence ratio can speed up the combustion velocity and improve the gas exchange process at low engine speeds of turbocharged hydrogen engines.
(3) The cycle number more than 150 can ensure that the variation amplitude of cumulative average is smaller than 0.3% for turbocharged hydrogen engines at low engine speeds. The COV_{imep} of turbocharged hydrogen is lower than 1.5% and decrease with increase of the equivalence ratio.
(4) The NO_x emission can be reduced by the HC, CO and H_2 under oxygen-rich environment but the conversion efficiency is very low. When the equivalence ratio is bigger than 0.95, the NO_x emission at behind of TWC presents a cliff-like decline and the value is close to 0 because the oxygen concentration decreases and the H_2 can be used to reduce NO_x in the TWC.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2019.03.245.

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